

LAB ACTION PLAN FOR WEEK 6

Objectives:

Students will be able to:

- Learn how to define and run multiple interdependent services (e.g., web server, database) in a single configuration file.
- Gain skills in writing and interpreting docker-compose.yml files for service setup.
- Deploy the same setup across different machines without manual configuration.
- Configure container networking and persistent storage within Compose.
- Reduce setup time and enable faster iteration during application development.

Students will perform hands-on running multi-container applications using docker compose and make a document that includes the execution of each step along with scenario-based question solutions after completing the tasks.

What is Docker Compose?

Docker Compose is a tool used to define and run multi-container Docker applications. It allows you to define services, networks, and volumes that your application needs, all in a single file. This makes it easier to manage complex applications that require multiple containers (e.g., a web server and a database).

Structure of a Docker Compose File

A Docker Compose file is written in YAML (YAML Ain't Markup Language), a human-readable data format used for configuration. In the Docker context, it helps define the services, networks, and volumes needed for an application.

Basic Components of a Docker Compose YAML File

1. **Version:** Specifies which version of the Docker Compose file format you're using. Each version comes with its own set of features.

```
version: '3.8'
```

2. **Services:** This section defines all the containers you want to run for your application. Each service represents a container (like a web server or database).

Each service includes:

- **Image:** The Docker image to use.
- **Ports:** Ports to map between the container and your host machine.
- **Environment:** Environment variables required by the service.
- **Dependencies:** Which other services a container depends on.

Example:

```
services:
web:
image:nginx
ports:
-"8060:80"
db:
image:tomee
ports:
-"8050:8080"
```

3. **Networks:** Define networks to allow different services to communicate with each other. Docker Compose automatically creates a default network if not specified.
4. **Volumes:** Used for persistent storage, allowing data to persist even after the container stops or is removed.

Example Docker Compose File (Simple)

Here's a basic example of a Docker Compose file that runs WordPress and MySQL together:

```
version: '3.8' # Docker Compose file format version

services:
wordpress: # WordPress service
image: wordpress:latest
ports:
- "8080:80" # Map port 80 of the container to port 8080 of the host
environment:
WORDPRESS_DB_HOST: db:3306 # Database host
WORDPRESS_DB_USER: wordpress
WORDPRESS_DB_PASSWORD: wordpress
WORDPRESS_DB_NAME: wordpress
depends_on:
```

- db # Ensures the db service starts first

```
db: # MySQL service
  image: mysql:5.7
  environment:
    MYSQL_ROOT_PASSWORD: rootpassword
    MYSQL_DATABASE: wordpress
    MYSQL_USER: wordpress
    MYSQL_PASSWORD: wordpress
```

How to Run It

To run a multi-container setup like the one above:

- 1. Save the file as docker-compose.yml.**

Or

docker-compose.yaml

- 2. To Start the compose**

docker-compose up -d

- 3. To stop the containers**

docker-compose down

- 4. To scale the container**

docker-compose up --scale<service name>=2 -d

1. Define and run multiple interdependent services

Task:

- I. Create a new folder compose-lab**

Inside it, create a file docker-compose.yml with the following content:

```
version: "3.9"
```

```
services:
```

```
  web:
```

image: nginx:latest

ports:

- "8080:80"

db:

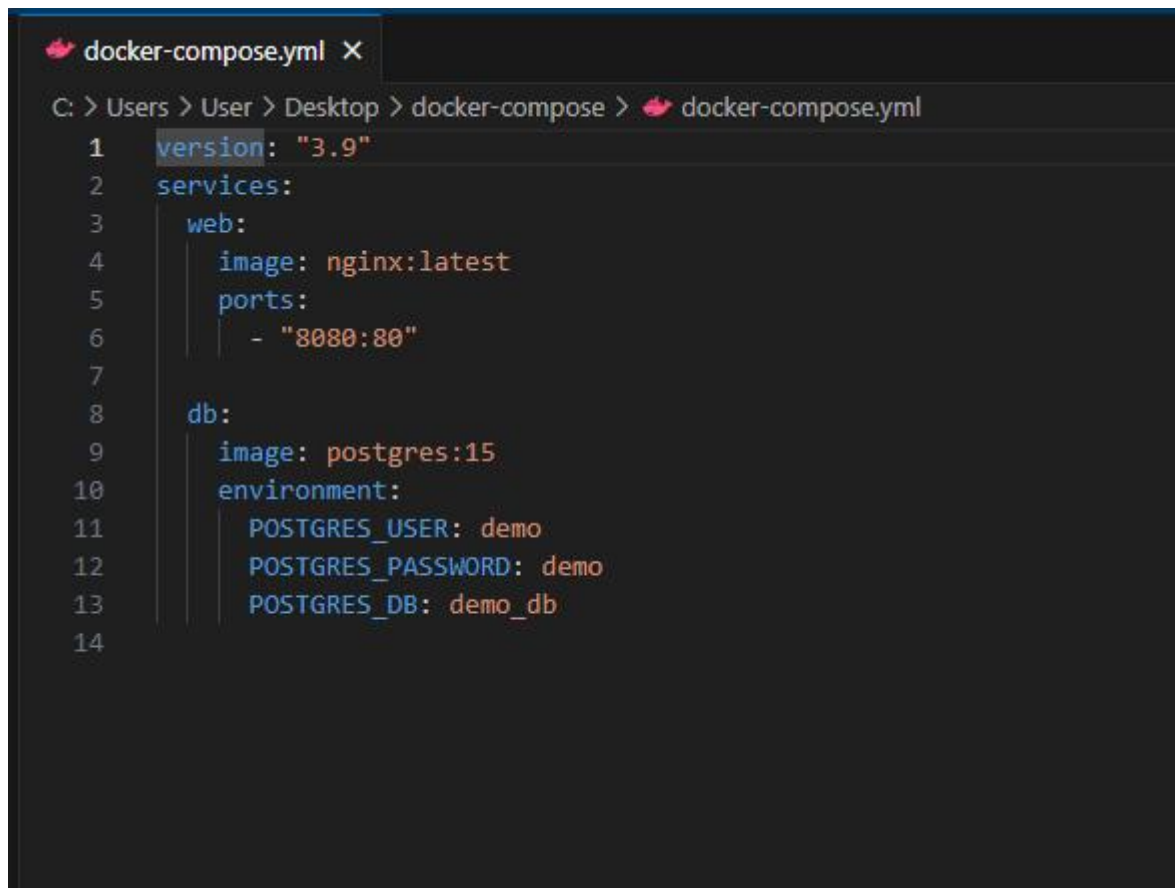
image: postgres:15

environment:

POSTGRES_USER: demo

POSTGRES_PASSWORD: demo

POSTGRES_DB: demo_db



```
docker-compose.yml X
C: > Users > User > Desktop > docker-compose > docker-compose.yml
1  version: "3.9"
2  services:
3    web:
4      image: nginx:latest
5      ports:
6        - "8080:80"
7
8    db:
9      image: postgres:15
10     environment:
11       POSTGRES_USER: demo
12       POSTGRES_PASSWORD: demo
13       POSTGRES_DB: demo_db
14
```

II.Run the setup:

docker compose up -d

III.Open your browser and visit: <http://localhost:8080>.

IV.Expected Output:

Nginx welcome page is displayed.

db container runs in the background.



2.Write and interpret docker-compose.yml files

Task:

I. Modify docker-compose.yml to add a Redis cache:

redis:

image: redis:alpine

II.Add a depends_on so web waits for Redis:

web:

image: nginx:latest

ports:

- "8080:80"

depends_on:

- redis

```
docker-compose.yml X
C: > Users > User > Desktop > docker-compose > docker-compose.yml
1  version: "3.9"
2  services:
3    web:
4      image: nginx:latest
5      ports:
6      - "8081:80"
7      depends_on:
8      - redis
9
10   redis:
11     image: redis:alpine
12
13   db:
14     image: postgres:15
15     environment:
16       POSTGRES_USER: demo
17       POSTGRES_PASSWORD: demo
18       POSTGRES_DB: demo_db
19
```

III.Restart the setup:

docker compose up -d

```
C:\Users\User\Desktop\docker-compose>docker compose up -d
time="2025-08-26T11:28:34+05:30" level=warning msg="C:\\Users\\User\\Desktop\\docker-compose\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 4/4
  ✔ Network docker-compose_default      Created                                2.1s
  ✔ Container docker-compose-db-1       Started                                5.8s
  ✔ Container docker-compose-redis-1    Started                                0.0s
  ✔ Container docker-compose-web-1      Started                                4.0s
```

docker compose ps

```
C:\Users\User\Desktop\docker-compose>docker compose ps
time="2025-08-26T11:26:03+05:30" level=warning msg="C:\\Users\\User\\Desktop\\docker-compose\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 8/8
  ✔ redis Pulled                                12.6s
  ✔ 61cf050eeff3 Download complete              1.2s
  ✔ 160694ef5d62 Download complete              1.2s
  ✔ 1f79dac8d2d4 Download complete              1.2s
  ✔ 4f4fb700ef54 Already exists                 0.0s
  ✔ 9880d81fff87a Download complete             0.9s
  ✔ f8eab6d4856e Download complete             7.7s
  ✔ 9824c27679d3 Download complete             3.4s
[+] Running 3/3
  ✔ Container docker-compose-db-1      Running                                0.0s
  ✔ Container docker-compose-redis-1   Started                                3.6s
  ✔ Container docker-compose-web-1     Running                                0.0s

C:\Users\User\Desktop\docker-compose>docker compose ps
NAME                IMAGE             COMMAND                  SERVICE    CREATED         STATUS         PORTS
docker-compose-db-1 postgres:15        "docker-entrypoint.s..." db          7 minutes ago   Up 7 minutes   5432/tcp
docker-compose-redis-1 redis:alpine       "docker-entrypoint.s..." redis       28 seconds ago Up 25 seconds   6379/tcp
docker-compose-web-1 nginx:latest       "/docker-entrypoint...." web         2 minutes ago   Up 2 minutes   0.0.0.0:8081->80/tcp
```

IV.Expected Output:

Three services (web, db, redis) are listed as running.

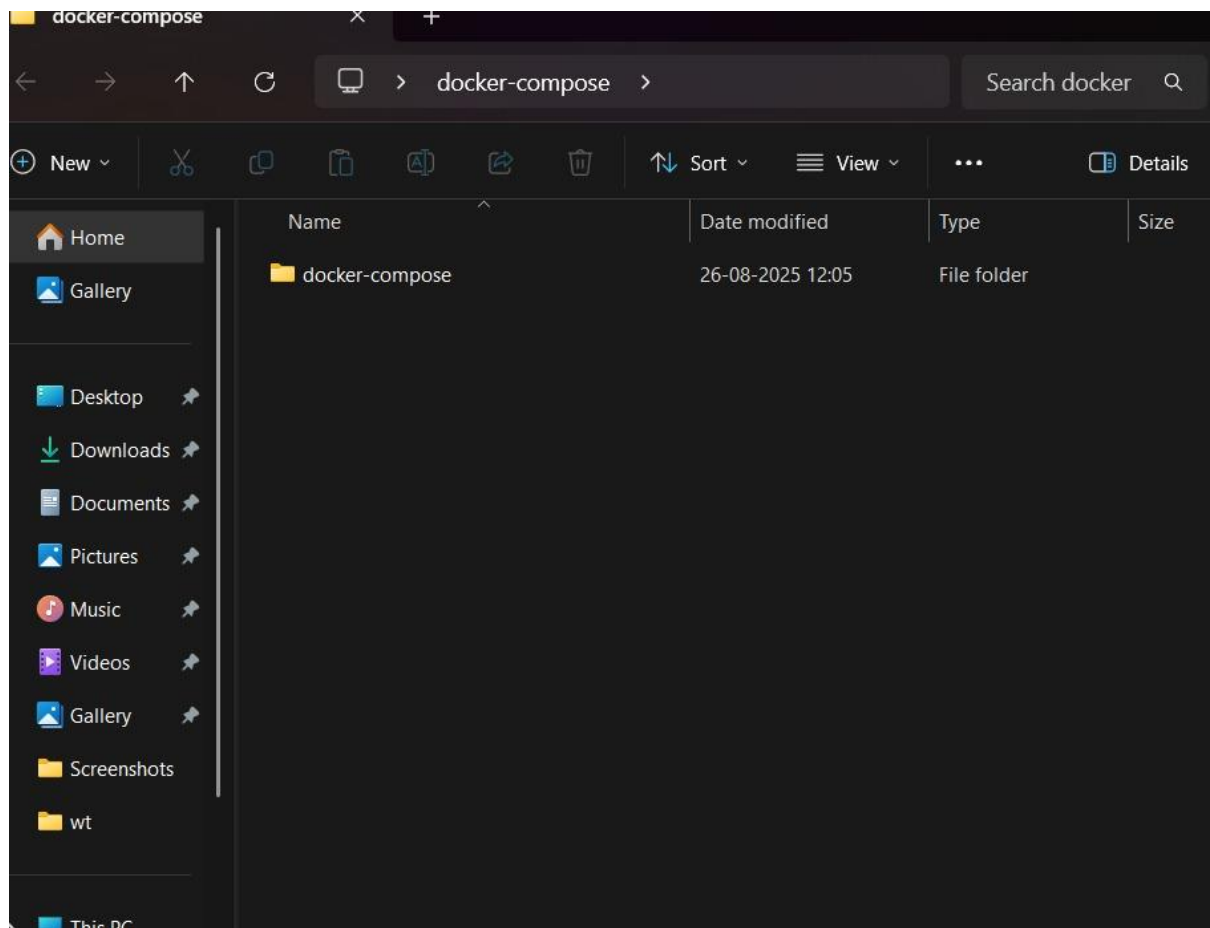
```
C:\Users\User\Desktop\docker-compose>docker compose ps
time="2025-08-26T11:28:53+05:30" level=warning msg="C:\\Users\\User\\Desktop\\docker-compose\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
docker-compose-db-1  postgres:15          "docker-entrypoint.s..." db         16 seconds ago Up 11 seconds  5432/tcp
docker-compose-redis-1  redis:alpine         "docker-entrypoint.s..." redis      16 seconds ago Up 11 seconds  6379/tcp
docker-compose-web-1  nginx:latest         "/docker-entrypoint...." web        13 seconds ago Up 9 seconds   0.0.0.0:8081->80/tcp
```

3.Deploy across different machines

Task:

I. Zip your compose-lab folder.

Transfer it to another machine with Docker Compose installed.



II.Run:

docker compose up -d

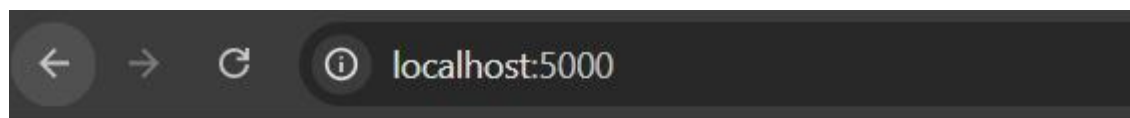
Check that Nginx and Postgres work there as well.

```
✓+5d4d246bc6e Pull complete 97.0s
✓ba7036bb0a60 Pull complete 97.0s
✓799f859abbbd Pull complete 97.0s
✓d903e777080c Pull complete 97.1s
✓04adcb462801 Pull complete 97.1s
✓5c10925b29ff Pull complete 97.1s
[+] Running 0/1.8s (4/8) docker:desktop-linux
[+] Running 0/1 Building 21.5s
[+] Building 27.4s (7/8) docker:desktop-linux
[+] Building 28.2s (10/10) FINISHED docker:desktop-linux
=> [web internal] load build definition from Dockerfile
=> => transferring dockerfile: 142B 0.0s
=> [web internal] load metadata for docker.io/library/python:3.10-slim 5.6s
=> [web internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [web 1/4] FROM docker.io/library/python:3.10-slim@sha256:420fbb0e468d3eaf0f7e93ea6f7a48792cbcad39d43ac95b96 16.2s
=> => resolve docker.io/library/python:3.10-slim@sha256:420fbb0e468d3eaf0f7e93ea6f7a48792cbcad39d43ac95b96bee2a 0.0s
=> => sha256:7732878f45d9e71f91ce50493915297ccalbde392445d9ddcc0f378a200967bf 1.29MB / 1.29MB 1.6s
=> => sha256:72e8e193aa94d19c7f1bcb00737a83d1906bcc1e51965c2873f081eb87bd3a0 14.10MB / 14.10MB 15.0s
=> => sha256:3a195ff1e16155a2ca71eee2cc2c4e467119c644d0360b7c2f6e6d9633f9358b 250B / 250B 1.7s
=> => sha256:420fbb0e468d3eaf0f7e93ea6f7a48792cbcad39d43ac95b96bee2afe4367da 10.37kB / 10.37kB 0.0s
=> => sha256:e3c437eb1623e9086f42038c1273a7eff09cfff44d7c2ece721fa61560af7e1f 1.75kB / 1.75kB 0.0s
=> => sha256:c2c9cfb78b6afd7fa4f08488dabdd4482417fa3bbc3571425c20b53d66636fc0 5.38kB / 5.38kB 0.0s
=> => extracting sha256:7732878f45d9e71f91ce50493915297ccalbde392445d9ddcc0f378a200967bf 0.2s
=> => extracting sha256:72e8e193aa94d19c7f1bcb00737a83d1906bcc1e51965c2873f081eb87bd3a0 1.1s
=> => extracting sha256:3a195ff1e16155a2ca71eee2cc2c4e467119c644d0360b7c2f6e6d9633f9358b 0.0s
=> [web internal] load build context 0.0s
=> => transferring context: 221B 0.0s
=> [web 2/4] WORKDIR /app 0.2s
=> [web 3/4] COPY app.py /app/ 0.0s
=> [web 4/4] RUN pip install flask 5.8s
=> [web] exporting to image 0.2s
=> => exporting layers 0.1s
=> => writing image sha256:4fda6735cc10585ef46a42c8b6a2c9806942cbc441dbc207e5e3301422f92ebd 0.0s
[+] Running 6/60 docker.io/library/docker-compose-web 0.0s
✓Service web Built 28.5s
✓Network docker-compose_app-net Created 0.1s
✓Network docker-compose_default Created 0.0s
✓Volume "docker-compose_db-data" Created 0.0s
✓Container docker-compose-db-1 Started 0.6s
✓Container docker-compose-web-1 Started 1.0s
```

III.Expected Output:

The same services run on the new machine without changes.

```
C:\Users\ranga\OneDrive\Desktop\docker-compose\docker-compose>docker compose ps
NAME                IMAGE                COMMAND                  SERVICE    CREATED        STATUS        PORTS
docker-compose-db-1 postgres:15          "docker-entrypoint.s..." db         About a minute ago Up About a minute 5432/tcp
docker-compose-web-1 docker-compose-web   "python app.py"         web        About a minute ago Up About a minute 0.0.0.0:5000->5000/tcp
```



Hello Docker Compose!

4.Networking and persistent storage

Task:

- I. Update your docker-compose.yml to add a custom network and volume:

networks:

app-net:

volumes:

db-data:

services:

web:

image: nginx:latest

ports:

- "8080:80"

networks:

- app-net

depends_on:

- db

db:

image: postgres:15

environment:

POSTGRES_USER: demo

POSTGRES_PASSWORD: demo

POSTGRES_DB: demo_db

volumes:

- db-data:/var/lib/postgresql/data

networks:

- app-net

```
docker-compose.yml
C: > Users > User > Desktop > docker-compose > docker-compose.yml

1  services:
2    web:
3      image: nginx:latest
4      ports:
5        - "8081:80"
6      networks:
7        - app-net
8      depends_on:
9        - db
10
11   db:
12     image: postgres:15
13     environment:
14       POSTGRES_USER: demo
15       POSTGRES_PASSWORD: demo
16       POSTGRES_DB: demo_db
17     volumes:
18       - db-data:/var/lib/postgresql/data
19     networks:
20       - app-net
21
22   networks:
23     app-net:
24
25   volumes:
26     db-data:
27
```

II.Run:

docker compose up -d

```
[+] Running 0/1desktop\docker-compose>docker compose up -d
- Network docker-compose_app-net Creating 0.3s
[+] Running 4/4611:36:37+05:30" level=warning msg="Found orphan containers ([docker-compose-redis-1]) for this project. If you removed or renamed this service in your
- Network docker-compose_app-net Created 0.3s
- Volume "docker-compose_db-data" Created 0.0s
- Container docker-compose-web-1 Started 0.7s
- Container docker-compose-db-1 Started 3.1s
```

III.Insert some data into Postgres (optional with psql).

```
C:\Users\User\Desktop\docker-compose>docker compose exec db psql -U demo -d demo_db
psql (15.14 (Debian 15.14-1.pgdg13+1))
Type "help" for help.

demo_db=# CREATE TABLE test (id SERIAL PRIMARY KEY, name VARCHAR(50));
CREATE TABLE
demo_db=# INSERT INTO test (name) VALUES ('Sample data');
INSERT 0 1
demo_db=# SELECT * FROM test;
 id |      name
-----+-----
  1 | Sample data
(1 row)

demo_db=# \q
```

IV.Remove containers:

docker compose down

```
[+] Running 3/3esktop\docker-compose>docker compose down
[+] Container docker-compose-web-1 Removed
[+] Container docker-compose-db-1 Removed
[+] Network docker-compose_app-net Removed
```

V.Start again:

docker compose up -d

```
C:\Users\User\Desktop\docker-compose>docker compose up -d
[+] Running 0/0
- Network docker-compose_app-net Creating 0.0s
[+] Running 3/36T11:41:36+05:30" level=warning msg="Found orphan containers ([docker-compose-redis-1]) for this project. If you removed or renamed this service in your
[+] Network docker-compose_app-net Created 0.1s
[+] Container docker-compose-db-1 Started 1.0s
[+] Container docker-compose-web-1 Started 1.3s
```

VI.Expected Output:

Database data persists across restarts.

Services communicate via the app-net network using service names.

```
C:\Users\User\Desktop\docker-compose>docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
docker-compose-db-1 postgres:15          "docker-entrypoint.s..." db         About a minute ago Up About a minute 5432/tcp
docker-compose-redis-1 redis:alpine         "docker-entrypoint.s..." redis      14 minutes ago Up 14 minutes 6379/tcp
docker-compose-web-1 nginx:latest         "/docker-entrypoint..." web        About a minute ago Up About a minute 0.0.0.0:8081->80/tcp
```

5.Faster iteration during development

Task:

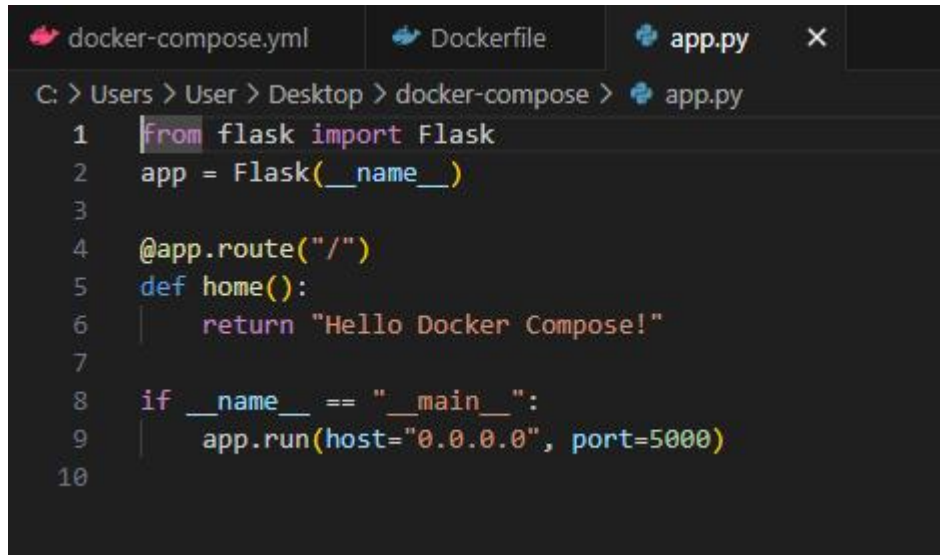
I. Create a simple Flask app in app.py:

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route("/")
```

```
def home():  
    return "Hello from Flask + Docker!"  
  
if __name__ == "__main__":  
    app.run(host="0.0.0.0", port=5000)
```

A screenshot of a code editor with three tabs: 'docker-compose.yml', 'Dockerfile', and 'app.py'. The 'app.py' tab is active, showing a Python script. The path in the editor is 'C: > Users > User > Desktop > docker-compose > app.py'. The code in 'app.py' is:

```
1 from flask import Flask  
2 app = Flask(__name__)  
3  
4 @app.route("/")  
5 def home():  
6     return "Hello Docker Compose!"  
7  
8 if __name__ == "__main__":  
9     app.run(host="0.0.0.0", port=5000)  
10
```

II.Add a Dockerfile in the same folder:

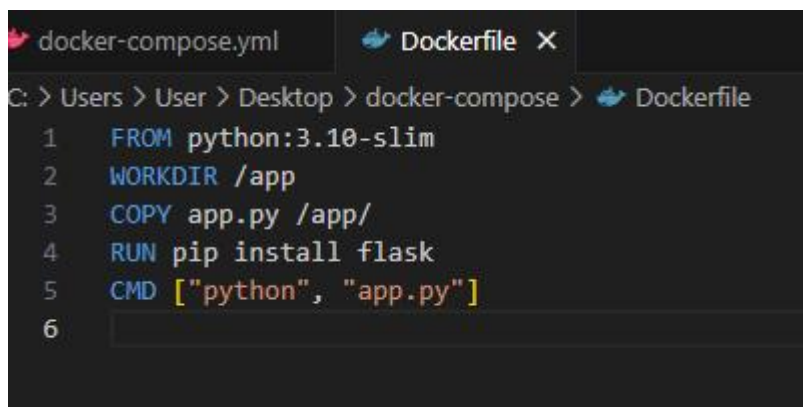
FROM python:3.10-slim

WORKDIR /app

COPY app.py /app/

RUN pip install flask

CMD ["python", "app.py"]

A screenshot of a code editor with two tabs: 'docker-compose.yml' and 'Dockerfile'. The 'Dockerfile' tab is active, showing a Dockerfile. The path in the editor is 'C: > Users > User > Desktop > docker-compose > Dockerfile'. The code in 'Dockerfile' is:

```
1 FROM python:3.10-slim  
2 WORKDIR /app  
3 COPY app.py /app/  
4 RUN pip install flask  
5 CMD ["python", "app.py"]  
6
```

III.Update docker-compose.yml:

web:

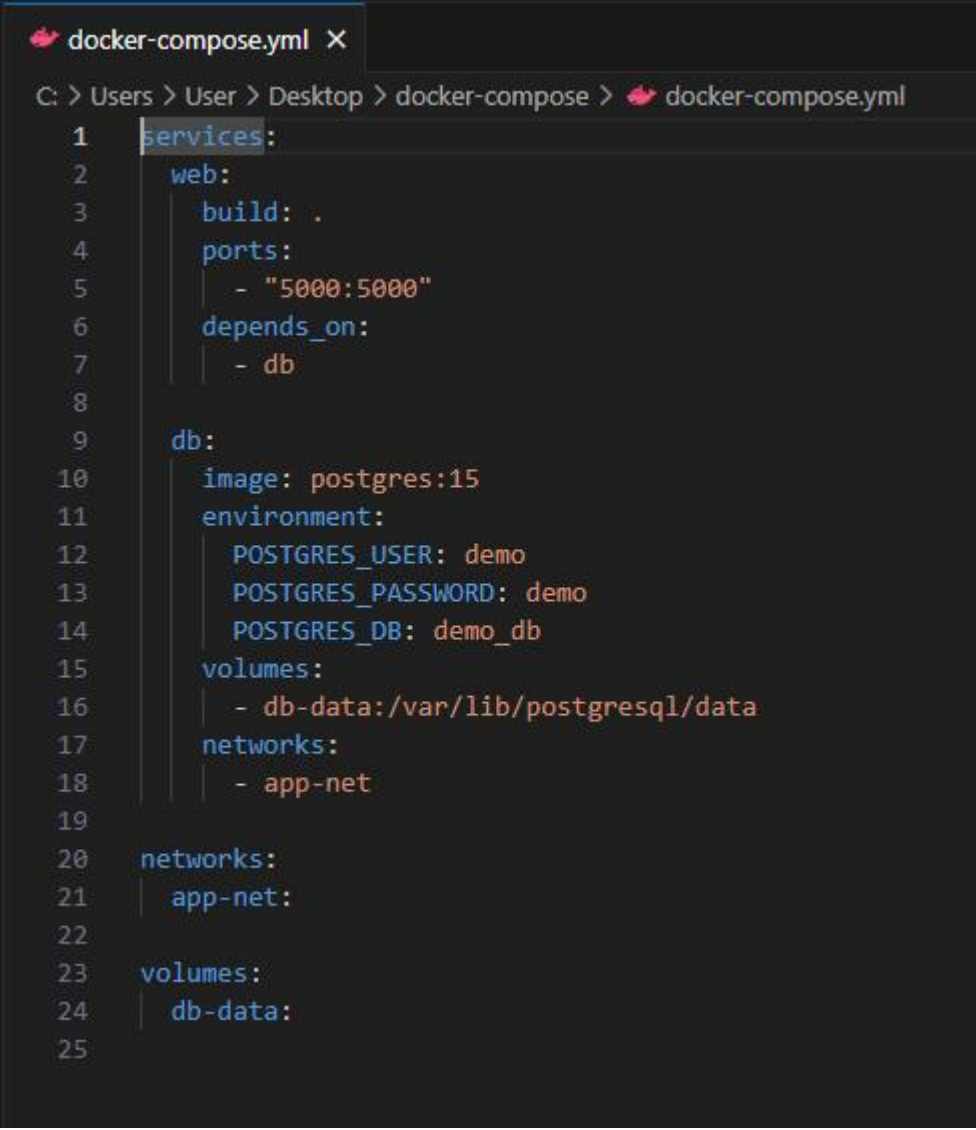
build: .

ports:

- "5000:5000"

depends_on:

- db

A screenshot of a code editor window titled 'docker-compose.yml'. The editor shows the following YAML configuration:

```
1 services:
2   web:
3     build: .
4     ports:
5       - "5000:5000"
6     depends_on:
7       - db
8
9   db:
10    image: postgres:15
11    environment:
12      POSTGRES_USER: demo
13      POSTGRES_PASSWORD: demo
14      POSTGRES_DB: demo_db
15    volumes:
16      - db-data:/var/lib/postgresql/data
17    networks:
18      - app-net
19
20 networks:
21   app-net:
22
23 volumes:
24   db-data:
```

The editor interface includes a file explorer on the left showing the path 'C: > Users > User > Desktop > docker-compose > docker-compose.yml' and a line number indicator on the left side of the code.

IV.Run:

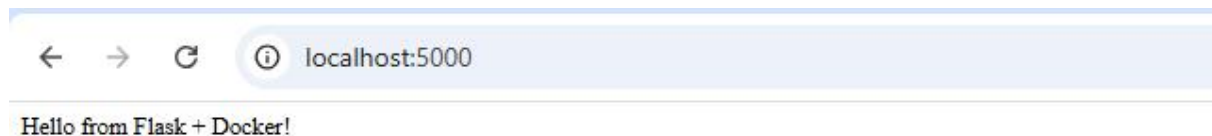
docker compose up --build

```

=> => transferring dockerfile: 142B 0.1s
=> [web internal] load metadata for docker.io/library/python:3.10-slim 3.2s
=> [web auth] library/python:pull token for registry-1.docker.io 0.0s
=> [web internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [web 1/4] FROM docker.io/library/python:3.10-slim@sha256:420fbb0e468d3eaf0f7e93ea6f7a48792cbcadc39d43ac95b96bee2afe4367da 7.5s
=> => resolve docker.io/library/python:3.10-slim@sha256:420fbb0e468d3eaf0f7e93ea6f7a48792cbcadc39d43ac95b96bee2afe4367da 0.0s
=> => sha256:3a195ff1e16155a2ca71ee2cc2c4e467119c644d0360b7c2f6e6d9633f9358b 250B / 250B 0.6s
=> => sha256:72e8e193aa94d19c7f1bcb00737a83d1906b0cc1e51965c2873f081eb87bd3a0 14.10MB / 14.10MB 4.0s
=> => sha256:7732878f45d9e71f91ce50493915297cca1bde392445d9ddcc0f378a2009e7bf 1.29MB / 1.29MB 1.3s
=> => extracting sha256:7732878f45d9e71f91ce50493915297cca1bde392445d9ddcc0f378a2009e7bf 0.4s
=> => extracting sha256:72e8e193aa94d19c7f1bcb00737a83d1906b0cc1e51965c2873f081eb87bd3a0 2.8s
=> => extracting sha256:3a195ff1e16155a2ca71ee2cc2c4e467119c644d0360b7c2f6e6d9633f9358b 0.4s
=> [web internal] load build context 0.2s
=> => transferring context: 226B 0.0s
=> [web 2/4] WORKDIR /app 3.2s
=> [web 3/4] COPY app.py /app/ 0.7s
=> [web 4/4] RUN pip install flask 6.7s
=> [web] exporting to image 4.3s
=> => exporting layers 3.3s
=> => exporting manifest sha256:611885e5d0b7a8a07e7fb024b7da7c5ecf0383cfb197f79556e9b061c916aaf7 0.0s
=> => exporting config sha256:1aa3e4986593159f47dbc581899b1be8d3894033c85723dd1e8b3a7e40a914d8 0.1s
=> => exporting attestation manifest sha256:56476c6140df2fecf038b4449f4fb14f005f59a641605edc433acc7f39963625 0.1s
=> => exporting manifest list sha256:c69d811fc20b802bb56f67dfc223e08fd887a78e0bdf39ff77be808512f6af3a9 0.0s
=> => naming to docker.io/library/docker-compose-web:latest 0.0s
=> => unpacking to docker.io/library/docker-compose-web:latest 0.7s
=> [web] resolving provenance for metadata file 0.1s
time="2025-08-26T11:48:52+05:30" level=warning msg="Found orphan containers ([docker-compose-redis-1]) for this project. If you removed or renamed this service in your compose file, you can run this command with the --remove-orphans flag to clean it up."
[+] Running 2/2
  Container docker-compose-db-1 Running 0.0s
  Container docker-compose-web-1 Recreated 1.7s
Attaching to db-1, web-1
web-1 | * Serving Flask app 'app'
web-1 | * Debug mode: off
web-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
web-1 | * Running on all addresses (0.0.0.0)
web-1 | * Running on http://172.18.0.1:5000
web-1 | * Running on http://172.18.0.3:5000
web-1 | Press CTRL+C to quit
web-1 | 172.18.0.1 - - [26/Aug/2025 06:19:12] "GET / HTTP/1.1" 200 -
web-1 | 172.18.0.1 - - [26/Aug/2025 06:19:13] "GET /favicon.ico HTTP/1.1" 404 -

```

Visit <http://localhost:5000>.



Change the return text in app.py (e.g., "Hello Docker Compose!").

V.Rebuild:

docker compose up –build

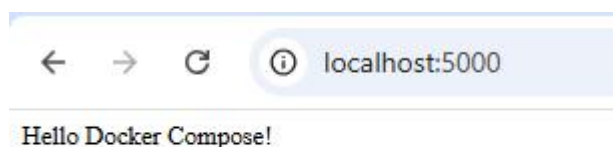

```

=> [web 4/4] RUN pip install flask 4.9s
=> [web] exporting to image 2.1s
=> => exporting layers 1.4s
=> => exporting manifest sha256:11d8ef7511aaa6dd3587131b80a434b055db49f1b26a6817360da44699f64ad9 0.0s
=> => exporting config sha256:02c7900a65747ba3b1b05717ead022da6e5be2ad2c5ff93b378e09c9c0a21cd8f 0.0s
=> => exporting attestation manifest sha256:2e29cf9ae4a50d5ec28a80bc28a834f3e561d26e8aff85dc99007d0d66f0903 0.0s
=> => exporting manifest list sha256:79c541e967f32d0d51131c7a65d48f2a63330ed01620506ae70264b31e326deb 0.0s
=> => naming to docker.io/library/docker-compose-web:latest 0.0s
=> => unpacking to docker.io/library/docker-compose-web:latest 0.6s
[web] resolving provenance for metadata file 0.0s
time="2025-08-26T11:51:07+05:30" level=warning msg="Found orphan containers ([docker-compose-redis-1]) for this project. If you removed or renamed this service in your compose file, you can run this command with the --remove-orphans flag to clean it up."
[*] Running 2/2
[+] Container docker-compose-db-1 Created 0.0s
[+] Container docker-compose-web-1 Recreated 0.8s
Attaching to db-1, web-1
db-1 | PostgreSQL Database directory appears to contain a database; Skipping initialization
db-1 |
db-1 | 2025-08-26 06:21:09.561 UTC [1] LOG: starting PostgreSQL 15.14 (Debian 15.14-1.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
db-1 | 2025-08-26 06:21:09.561 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
db-1 | 2025-08-26 06:21:09.561 UTC [1] LOG: listening on IPv6 address ":::", port 5432
db-1 | 2025-08-26 06:21:09.579 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
db-1 | 2025-08-26 06:21:09.606 UTC [20] LOG: database system was interrupted; last known up at 2025-08-26 06:16:37 UTC
db-1 | 2025-08-26 06:21:10.003 UTC [20] LOG: database system was not properly shut down; automatic recovery in progress
db-1 | 2025-08-26 06:21:10.071 UTC [20] LOG: redo starts at 0/198FE70
db-1 | 2025-08-26 06:21:10.071 UTC [20] LOG: invalid record length at 0/198FF58: wanted 24, got 0
db-1 | 2025-08-26 06:21:10.071 UTC [20] LOG: redo done at 0/198FF20 system usage: CPU: user: 0.00 s, system: 0.00 s, elapsed: 0.00 s
db-1 | 2025-08-26 06:21:10.082 UTC [27] LOG: checkpoint starting: end-of-recovery immediate wait
db-1 | 2025-08-26 06:21:10.111 UTC [27] LOG: checkpoint complete: wrote 3 buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.008 s, sync=0.006 s, total=0.033 s; sync files=2, longest=0.005 s, average=0.003 s; distance=0 kB, estimate=0 kB
db-1 | 2025-08-26 06:21:10.116 UTC [1] LOG: database system is ready to accept connections
web-1 | * Serving Flask app 'app'
web-1 | * Debug mode: off
web-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
web-1 | * Running on all addresses (0.0.0.0)
web-1 | * Running on http://127.0.0.1:5000
web-1 | * Running on http://172.18.0.3:5000
web-1 | Press CTRL+C to quit
web-1 | 172.18.0.1 - - [26/Aug/2025 06:21:13] "GET / HTTP/1.1" 200 -

```

VI.Expected Output:

New message appears instantly after rebuild.



Conclusion:

Docker Compose simplifies multi-container application management by allowing developers to define, configure, and run services in a single YAML file. It streamlines deployment, ensures consistent environments, and reduces manual setup, making it an essential tool for orchestrating complex, container-based applications efficiently.